

Part II free fixed-coarse rate

An explicit $D4$ filter estimate and covariance-symbol proof

Lamport-form proof

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Quantitative free theorem. Fix a coarse scale M , a torus $\mathbb{T}_L$, and a coarse Weyl generator $W_M(\varepsilon_M q + ip)$. For the MMST nearest-neighbour free vacua and the exact Daubechies $D4$ observable embeddings,

$$\left| \omega_{M+r,L}^{(0)}\left(\alpha_M^{M+r}(W_M(\varepsilon_M q + ip))\right) - \omega_L^{(0)}(\beta_M(W_M(\varepsilon_M q + ip))) \right| \leq C_{M,L,m,q,p} 2^{-\theta r}, \quad (\text{R})$$

for all sufficiently large r , where

$$\eta = 2(1 - \log_2 \sqrt{3}) > 0, \quad \theta = \frac{2\eta}{5 + \eta} > 0. \quad (\text{E})$$

Every constant is displayed in Section 4. The theorem holds in the correct fixed-coarse topology and makes no diagonal $M = N$ assertion.

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1 Exact covariance formulas

Fix M , and put $N = M + r$. Use all conventions from the companion model sheet. Write

$$\Gamma_\infty := \frac{\pi}{L}\mathbb{Z}, \quad \gamma(k) := \sqrt{m^2 + k^2}, \quad \gamma_N(k) := \sqrt{m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2)}. \quad (1.1)$$

Extend the coarse Fourier transforms $\widehat{q}_M, \widehat{p}_M$ periodically from Γ_M to Γ_∞ , with period $2\pi/\varepsilon_M$.

Define the finite and infinite mask products

$$P_r(\ell) := \prod_{j=1}^r m_0(2^{-j}\ell), \quad P_\infty(\ell) := \prod_{j=1}^\infty m_0(2^{-j}\ell) = \widehat{\phi}(\ell). \quad (1.2)$$

The equality in (1.2) is the Fourier form of the $D4$ scaling equation with $\widehat{\phi}(0) = 1$.

Lemma 1.1 (pulled-back fine covariance). *For $\xi = \varepsilon_M q + ip$,*

$$\omega_{M+r,L}^{(0)}(\alpha_M^{M+r}(W_M(\xi))) = e^{-Q_{r,M}(q,p)/4}, \quad (1.3)$$

where

$$Q_{r,M}(q,p) = \frac{\varepsilon_M}{2L} \sum_{k \in \Gamma_\infty} \mathbf{1}_{\Gamma_{M+r}}(k) |P_r(\varepsilon_M k)|^2 \left(\gamma_{M+r}(k)^{-1} |\widehat{q}_M(k)|^2 + \gamma_{M+r}(k) |\widehat{p}_M(k)|^2 \right). \quad (1.4)$$

Proof in Lamport form

⟨1⟩ 1 Iterate the momentum-space refinement map.

The one-step MMST map is multiplication by $\sqrt{2} m_0(\varepsilon_{j+1} k)$, together with periodic extension of the coarse Fourier vector. Therefore

$$\widehat{R_M^{M+r}}(q,p)(k) = 2^{r/2} P_r(\varepsilon_M k)(\widehat{q}_M(k), \widehat{p}_M(k)), \quad k \in \Gamma_{M+r}. \quad (1.5)$$

⟨1⟩ 2 Insert (1.5) in the fine vacuum covariance.

The fine normalized momentum sum has weight

$$\frac{1}{2r_{M+r}} = \frac{2^{-r}}{2r_M} = 2^{-r} \frac{\varepsilon_M}{2L}. \quad (1.6)$$

The squared factor $2^{r/2}$ in (1.5) cancels the 2^{-r} in (1.6). Substitution in the exact free state formula gives (1.3)–(1.4). $\square$

Lemma 1.2 (continuum pullback covariance).

$$\omega_L^{(0)}(\beta_M(W_M(\xi))) = e^{-Q_{\infty,M}(q,p)/4}, \quad (1.7)$$

where

$$Q_{\infty,M}(q,p) = \frac{\varepsilon_M}{2L} \sum_{k \in \Gamma_\infty} |P_\infty(\varepsilon_M k)|^2 \left(\gamma(k)^{-1} |\widehat{q}_M(k)|^2 + \gamma(k) |\widehat{p}_M(k)|^2 \right). \quad (1.8)$$

Proof. The continuum embedding sends q, p to linear combinations of the periodized scaling functions. Their Fourier coefficients equal $\varepsilon_M^{1/2} e^{-ikx} \widehat{\phi}(\varepsilon_M k)$. Parseval's identity on $\mathbb{T}_L$, followed by the continuum free-vacuum formula, gives (1.8). Equation (1.2) then identifies the multiplier with P_∞ . $\square$

2 A uniform algebraic envelope for the exact D4 product

Recall the exact factorization

$$m_0(t) = \left(\frac{1 + e^{-it}}{2} \right)^2 D(t), \quad D(t) = \frac{(1 + \sqrt{3}) + (1 - \sqrt{3})e^{-it}}{2}, \quad (2.1)$$

and

$$|D(t)|^2 = 2 - \cos t. \quad (2.2)$$

Set

$$a := \log_2 \sqrt{3} < 1, \quad \eta := 2 - 2a = 2(1 - \log_2 \sqrt{3}) > 0, \quad C_D := 48\pi^4 e^{1/6}. \quad (2.3)$$

Lemma 2.1 (uniform finite-product decay). *For every $r \geq 0$ and every ℓ satisfying $|\ell| \leq \pi 2^r$,*

$$|P_r(\ell)|^2 \leq C_D (1 + |\ell|)^{-2-\eta}. \quad (2.4)$$

The same estimate holds for P_∞ on $\mathbb{R}$.

Proof in Lamport form

⟨1⟩ 1 Bound the D -factor product.

Let

$$n(\ell) := \lceil \log_2 \max(1, |\ell|) \rceil. \quad (2.5)$$

By (2.2), $|D(t)| \leq \sqrt{3}$. Also

$$|D(t)|^2 = 1 + (1 - \cos t) \leq e^{t^2/2}, \quad |D(t)| \leq e^{t^2/4}. \quad (2.6)$$

For $j > n(\ell)$, use (2.6); for $j \leq n(\ell)$, use $|D| \leq \sqrt{3}$. Since

$$\sum_{j=n+1}^{\infty} 4^{-j} = \frac{4^{-n}}{3}, \quad |\ell|^2 4^{-n(\ell)} \leq 1, \quad (2.7)$$

we obtain, for every finite r ,

$$\prod_{j=1}^r |D(2^{-j}\ell)| \leq (\sqrt{3})^{n(\ell)} \exp\left(\frac{\ell^2}{4} \sum_{j=n(\ell)+1}^{\infty} 4^{-j}\right) \quad (1)$$

$$\leq \sqrt{3} e^{1/12} (1 + |\ell|)^a. \quad (2.8)$$

If $r < n(\ell)$, the omitted tail factor in the first line is absent and the same upper bound remains valid.

⟨1⟩ 2 Bound the cosine product.

For $r \geq 1$, the finite sine-product identity gives

$$\prod_{j=1}^r \left| \cos\left(\frac{\ell}{2^{j+1}}\right) \right| = \frac{|\sin(\ell/2)|}{2^r |\sin(\ell/2^{r+1})|}. \quad (2.9)$$

When $|\ell| \leq \pi 2^r$, the denominator angle has absolute value at most $\pi/2$. The elementary bound $|\sin u| \geq 2|u|/\pi$ on that interval yields

$$\prod_{j=1}^r \left| \cos\left(\frac{\ell}{2^{j+1}}\right) \right| \leq \min\left(1, \frac{\pi |\sin(\ell/2)|}{|\ell|}\right). \quad (2.10)$$

⟨1⟩ 3 Combine the factors.

Equations (2.1), (2.8), and (2.10) imply

$$|P_r(\ell)|^2 \leq 3e^{1/6} (1 + |\ell|)^{2a} \min(1, \pi^4 |\ell|^{-4}). \quad (2.11)$$

For $|\ell| \leq 1$, multiply the desired right side by $(1 + |\ell|)^4 \leq 16$. For $|\ell| \geq 1$, use $(1 + |\ell|)^4 |\ell|^{-4} \leq 16$. In both cases,

$$|P_r(\ell)|^2 \leq 48\pi^4 e^{1/6} (1 + |\ell|)^{-4+2a} = C_D (1 + |\ell|)^{-2-\eta}. \quad (2.12)$$

For $r = 0$, $P_0 = 1$ and $|\ell| \leq \pi$; the claimed bound follows from $C_D \geq (1 + \pi)^{2+\eta}$, which holds for $C_D = 48\pi^4 e^{1/6}$. Finally, take $r \rightarrow \infty$ in (2.12) and use (1.2) to obtain the infinite-product bound.

□

Lemma 2.2 (explicit filter-tail error). *If $|\ell| \leq 2^r$, then*

$$0 \leq |P_r(\ell)|^2 - |P_\infty(\ell)|^2 \leq \frac{\sqrt{3}}{12} \ell^2 4^{-r}. \quad (2.13)$$

Proof in Lamport form

⟨1⟩ 1 Compute the second derivative exactly.

Let $u(t) := |m_0(t)|^2$. From (2.1)–(2.2),

$$u(t) = \frac{2 + 3 \cos t - \cos^3 t}{4}, \quad u'(t) = -\frac{3}{4} \sin^3 t, \quad u''(t) = -\frac{9}{4} \sin^2 t \cos t. \quad (2.14)$$

The maximum of $x(1 - x^2)$ on $[0, 1]$ is $2/(3\sqrt{3})$, attained at $x = 1/\sqrt{3}$. Hence

$$\sup_t |u''(t)| = \frac{\sqrt{3}}{2}. \quad (2.15)$$

Because $u(0) = 1$ and $u'(0) = 0$, Taylor's theorem gives

$$0 \leq 1 - u(t) \leq \frac{\sqrt{3}}{4} t^2. \quad (2.16)$$

The left inequality follows independently from the QMF identity $u(t) + u(t + \pi) = 1$.

⟨1⟩ 2 Estimate the omitted infinite product.

Since every $u(t) \in [0, 1]$,

$$1 - \prod_{j=r+1}^{\infty} u(2^{-j}\ell) \leq \sum_{j=r+1}^{\infty} (1 - u(2^{-j}\ell)). \quad (2.17)$$

For $|\ell| \leq 2^r$, all arguments on the right have absolute value at most $1/2$. Equations (2.16)–(2.17) give

$$1 - \prod_{j=r+1}^{\infty} u(2^{-j}\ell) \leq \frac{\sqrt{3}}{4} \ell^2 \sum_{j=r+1}^{\infty} 4^{-j} = \frac{\sqrt{3}}{12} \ell^2 4^{-r}. \quad (2.18)$$

Multiplication by $|P_r(\ell)|^2 \leq 1$ proves (2.13). $\square$

3 Dispersion error

Lemma 3.1 (nearest-neighbour versus continuum symbol). *If $|\varepsilon_N k| \leq \pi$, then*

$$0 \leq \gamma(k) - \gamma_N(k) \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad (3.1)$$

$$0 \leq \gamma_N(k)^{-1} - \gamma(k)^{-1} \leq \frac{\varepsilon_N^2 k^4}{24m^3}. \quad (3.2)$$

Proof in Lamport form

⟨1⟩ 1 Bound the discrete Laplacian symbol.

For $0 \leq x \leq \pi/2$,

$$\sin x \leq x, \quad \sin x \geq x - \frac{x^3}{6}. \quad (3.3)$$

For the second inequality, the derivative of $\sin x - x + x^3/6$ is nonnegative because $\cos x \geq 1 - x^2/2$. Since $x - x^3/6 \geq 0$ on this interval,

$$x^2 - \frac{x^4}{3} \leq \left(x - \frac{x^3}{6}\right)^2 \leq \sin^2 x \leq x^2. \quad (3.4)$$

Putting $x = |\varepsilon_N k|/2$ gives

$$0 \leq k^2 - 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2) \leq \frac{\varepsilon_N^2 k^4}{12}. \quad (3.5)$$

⟨1⟩ 2 Pass to square roots and inverses.

Since both dispersions are at least m ,

$$\gamma - \gamma_N = \frac{\gamma^2 - \gamma_N^2}{\gamma + \gamma_N} \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad (3.6)$$

which is (3.1). Moreover,

$$\gamma_N^{-1} - \gamma^{-1} = \frac{\gamma - \gamma_N}{\gamma\gamma_N} \leq \frac{\varepsilon_N^2 k^4}{24m^3}, \quad (3.7)$$

which is (3.2). $\square$

4 The explicit rate

Put

$$A_q := \sup_{k \in \Gamma_\infty} |\widehat{q}_M(k)|, \quad A_p := \sup_{k \in \Gamma_\infty} |\widehat{p}_M(k)|, \quad (4.1)$$

which are finite because the extensions are periodic. Define

$$\kappa := \frac{\pi}{L}, \quad b := \varepsilon_M \kappa = \frac{\pi}{r_M}, \quad c_0 := \frac{\sqrt{3}}{12}. \quad (4.2)$$

Set

$$C_{\text{low}} := \frac{3\varepsilon_M^3}{2L} \left\{ A_q^2 \left(\frac{\kappa^4}{24m^3} + \frac{c_0 \kappa^2}{m} \right) \right. \quad (2)$$

$$\left. + A_p^2 \left(\frac{\kappa^4}{24m} + c_0(m + \kappa)\kappa^2 \right) \right\}, \quad (4.3)$$

$$C_* := 2C_D \left(\frac{A_q^2}{m} + A_p^2(m + \varepsilon_M^{-1}) \right). \quad (4.4)$$

Finally, let

$$\alpha := \frac{2}{5 + \eta}, \quad \theta := \alpha\eta = \frac{2\eta}{5 + \eta} = 2 - 5\alpha. \quad (4.5)$$

Theorem 4.1 (fixed-coarse power rate). *Let r_* be the least integer satisfying*

$$2^{\alpha r_*} \geq \max(1, 2b). \quad (4.6)$$

For every $r \geq r_$,*

$$|Q_{r,M}(q, p) - Q_{\infty,M}(q, p)| \leq C_Q 2^{-\theta r}, \quad (4.7)$$

where

$$C_Q := C_{\text{low}}(1 + b^{-1})^5 + \frac{2^\eta C_*}{\pi\eta}. \quad (4.8)$$

Consequently,

$$\left| \omega_{M+r,L}^{(0)}(\alpha_M^{M+r}(W_M(\varepsilon_M q + ip))) - \omega_L^{(0)}(\beta_M(W_M(\varepsilon_M q + ip))) \right| \leq \frac{C_Q}{4} 2^{-\theta r}. \quad (4.9)$$

Proof in Lamport form

⟨1⟩ 1 Choose an integer momentum split.

For $r \geq r_*$, put

$$J_r := \left\lfloor b^{-1} 2^{\alpha r} \right\rfloor. \quad (4.10)$$

Then $J_r \geq 1$,

$$bJ_r \leq 2^{\alpha r} \leq 2^r, \quad bJ_r \geq 2^{\alpha r} - b \geq \frac{1}{2} 2^{\alpha r}, \quad (4.11)$$

and every mode $k_n = \kappa n$ with $|n| \leq J_r$ lies in Γ_{M+r} . The first inequality in (4.11) also permits use of Lemma 2.2.

⟨1⟩ 2 Bound the low-momentum part.

For $|n| \leq J_r$, split the momentum integrand as

$$|\gamma_N |P_r|^2 - \gamma |P_\infty|^2| \leq |\gamma_N - \gamma| |P_r|^2 + \gamma ||P_r|^2 - |P_\infty|^2|, \quad (4.12)$$

$$|\gamma_N^{-1} |P_r|^2 - \gamma^{-1} |P_\infty|^2| \leq |\gamma_N^{-1} - \gamma^{-1}| |P_r|^2 + \gamma^{-1} ||P_r|^2 - |P_\infty|^2|. \quad (4.13)$$

Here every mask argument is $\ell = \varepsilon_M k_n$. Lemmas 2.2 and 3.1, together with $\varepsilon_{M+r} = 2^{-r} \varepsilon_M$, give

$$q\text{-integrand error} \leq 4^{-r} \varepsilon_M^2 A_q^2 \left(\frac{k_n^4}{24m^3} + \frac{c_0 k_n^2}{m} \right), \quad (4.14)$$

$$p\text{-integrand error} \leq 4^{-r} \varepsilon_M^2 A_p^2 \left(\frac{k_n^4}{24m} + c_0 \gamma(k_n) k_n^2 \right). \quad (4.15)$$

Because

$$2J + 1 \leq 3(1 + J), \quad |k_n| \leq \kappa(1 + J), \quad \gamma(k_n) \leq m + |k_n| \leq (m + \kappa)(1 + J), \quad (4.16)$$

the definition (4.3) yields

$$E_{\text{low}}(r) \leq 4^{-r} C_{\text{low}}(1 + J_r)^5 \leq C_{\text{low}}(1 + b^{-1})^5 2^{-(2-5\alpha)r}. \quad (4.17)$$

⟨1⟩ 3 Bound both high-momentum tails.

Lemma 2.1 and

$$\gamma_{M+r}(k) \leq \gamma(k) \leq m + |k| \leq (m + \varepsilon_M^{-1})(1 + \varepsilon_M |k|) \quad (4.18)$$

show that the sum of the absolute fine and continuum integrands at $k_n = \kappa n$ is at most

$$C_*(1 + b|n|)^{-1-\eta}. \quad (4.19)$$

The fine term is declared zero outside Γ_{M+r} , exactly as in (1.4). Since $x \mapsto (1 + bx)^{-1-\eta}$ is decreasing,

$$\sum_{|n| > J} (1 + b|n|)^{-1-\eta} \leq 2 \int_J^\infty (1 + bx)^{-1-\eta} dx = \frac{2}{b\eta} (1 + bJ)^{-\eta}. \quad (4.20)$$

Multiplication by the covariance weight $\varepsilon_M/(2L)$, followed by $b = \pi \varepsilon_M/L$ and (4.11), gives

$$E_{\text{high}}(r) \leq \frac{C_*}{\pi\eta} (1 + bJ_r)^{-\eta} \leq \frac{2^\eta C_*}{\pi\eta} 2^{-\alpha\eta r}. \quad (4.21)$$

⟨1⟩ 4 Match the exponents.

By (4.5),

$$2 - 5\alpha = \alpha\eta = \theta. \quad (4.22)$$

Adding (4.17) and (4.21) proves (4.7)–(4.8).

⟨1⟩ 5 Pass from covariance forms to states.

For $x, y \geq 0$, the mean-value theorem gives

$$|e^{-x/4} - e^{-y/4}| = \frac{1}{4}e^{-\zeta/4}|x - y| \leq \frac{1}{4}|x - y| \quad (4.23)$$

for some ζ between x and y . Apply (4.23) to $Q_{r,M}$ and $Q_{\infty,M}$ and use (4.7). This is (4.9). $\square$

Corollary 4.2 (lattice-spacing form). Because $\varepsilon_{M+r}/\varepsilon_M = 2^{-r}$, (4.9) is equivalently

$$|\omega_{M+r,L}^{(0)} \circ \alpha_M^{M+r}(W_M(\xi)) - \omega_L^{(0)} \circ \beta_M(W_M(\xi))| \leq \frac{C_Q}{4} \left(\frac{\varepsilon_{M+r}}{\varepsilon_M} \right)^\theta. \quad (4.24)$$

5 Claim ledger

Claim	Status	Exact reason
Fixed-coarse free convergence	PASS	The covariance formulas are (1.4) and (1.8), and Theorem 4.1 supplies a power rate.
Uniform finite $D4$ filter tail	PASS	Lemmas 2.1 and 2.2 prove both the global envelope and the low-frequency truncation error from the exact four mask coefficients.
Finest-scale local state norm	FAIL	This theorem fixes M and sends $r = N - M \rightarrow \infty$; it does not include $M = N$.
Interacting covariance reduction	FAIL	Interacting ground states are not Gaussian, so no quadratic covariance symbol determines their Weyl expectations.

References

- [1] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Commun. Math. Phys. **387** (2021), 1299–1367.